# Rushil Barve

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## Education

**University of Waterloo** — Waterloo, ON
*Bachelor of Computer Science (Co-op), Computational Mathematics* — *Expected 2031*

**Delhi Public School – Modern Indian School** — Doha, Qatar
*CBSE Senior Secondary, Computer Science* — *Graduated 2026*

## Projects

**CircuitGuard** | *Python, FastAPI, React, Three.js* — Sep 2026

- Real-time Formula 1 safety system: analyses live driver telemetry to detect crashes and stopped cars on the racing line and recommends the FIA response (safety car, VSC, red flag) under the 2026 sporting regulations.
- Models per-sector spectator risk with a Bayesian Monte Carlo insurance-pricing model, so premiums can reflect each track section instead of a blanket rate.
- Python/FastAPI backend on the OpenF1 API, FAISS semantic search over FIA regulations, faster-whisper radio transcription, Next.js/React 3D track view.

**Whisper Chain** | *Godot 4, GDScript* — Oct 2026

- Puzzle game where each villager garbles a message (reverse, swap, double, drop a word); players route the message so the distortions cancel, discovering inverses and non-commutativity.
- Graph solver (DFS over simple paths) finds every correct route, sets par and powers live hints; used to design and validate all 10 levels.
- ~3,000 lines of GDScript with UI-free game logic, ~1,500 test assertions including randomized property tests, and GitHub Actions CI building web and Windows exports.

**MUN Chair Toolkit** | *TypeScript, React, Vite, Tailwind, Zustand, Dexie* — Sep 2026

- Committee-management app for Model UN chairs: committee and agenda setup, bulk delegation entry, and roll call with Present, Present and Voting, and Absent states.
- Automated quorum, simple-majority and two-thirds calculations, so chairs no longer count votes by hand.

**Door Occupancy Counter (in progress)** | *Python, SQLite, JavaScript, TensorFlow.js* — Oct 2026 – Present

- Counts people entering and leaving each door from two-beam sensor events (direction from beam order), keeping a live headcount with no manual logging.
- Live web dashboard, SQLite event log and automatic daily Excel reports (summary, hourly and event sheets) written with no external libraries.

## Experience & Leadership

**Waterloo Student Design Teams** — Sep 2026 – Present
*Member: WARG, UW Orbital, UW Game Dev* — *Waterloo, ON*

- Waterloo Aerial Robotics Group (WARG): member of the Autonomy and Embedded Flight Software sub-teams.

**DPS-Modern Indian School** — May 2025 – Jun 2026
*Director of MUN & Electoral and Democratic Club* — *Doha, Qatar*

- Led the school's Model UN program: organized conferences, trained delegates in research, procedure and resolution drafting.

**TEDxYouth@DPSMIS** — Sep – Dec 2025
*Head of Marketing* — *Doha, Qatar*

- Led branding, social media and promotion for a TEDx-licensed youth event (11th edition, "Flip the Narrative").

**DPSDQ MUN (25th-anniversary edition)** — Jun – Oct 2025
*Head of Operations* — *Doha, Qatar*

- Ran logistics and delegate services for the conference, coordinating with the Secretariat and support teams.

## Hackathons & Awards

- 8 hackathons, 5 won, 1 organised: FormulaTech Hacks 2026 (Waterloo; Winner, Ampere: AI for Motor Safety), RevenueCat Shipaton, OpenCV AI Competition (AWS), Nebius x NVIDIA Global AI Hackathon, IBM BOB 2.0 (all 2026), DPS MIS Doha (2025), Birla School (2024), CMU Qatar (2023).
- Gold Medal of Distinction, International Informatics Olympiad (2023-24); Bronze (2024-25)
- Gold Medal of Distinction, National Science Olympiad
- Six Best Delegate awards across 8+ national and international MUN conferences
- Basketball, CBSE Qatar Clusters: Gold (2025-26), Bronze (2023-24, 2024-25)

## Technical Skills

**Languages**: Python, JavaScript, TypeScript, GDScript, C, SQL, HTML, CSS

**Frameworks**: React, Vite, Tailwind CSS, FastAPI, Node.js, Express, Godot 4, Zustand

**Tools**: Git, GitHub Actions, SQLite, IndexedDB, Render, Claude API, PayPal REST APIs